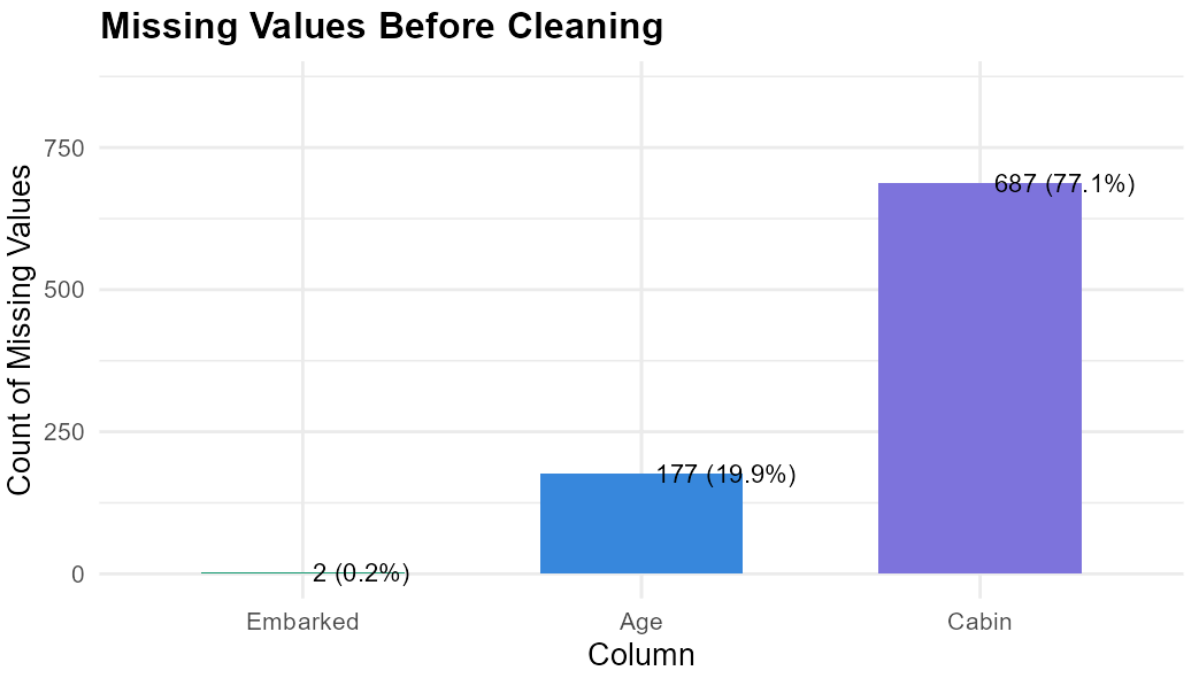
Lab 8

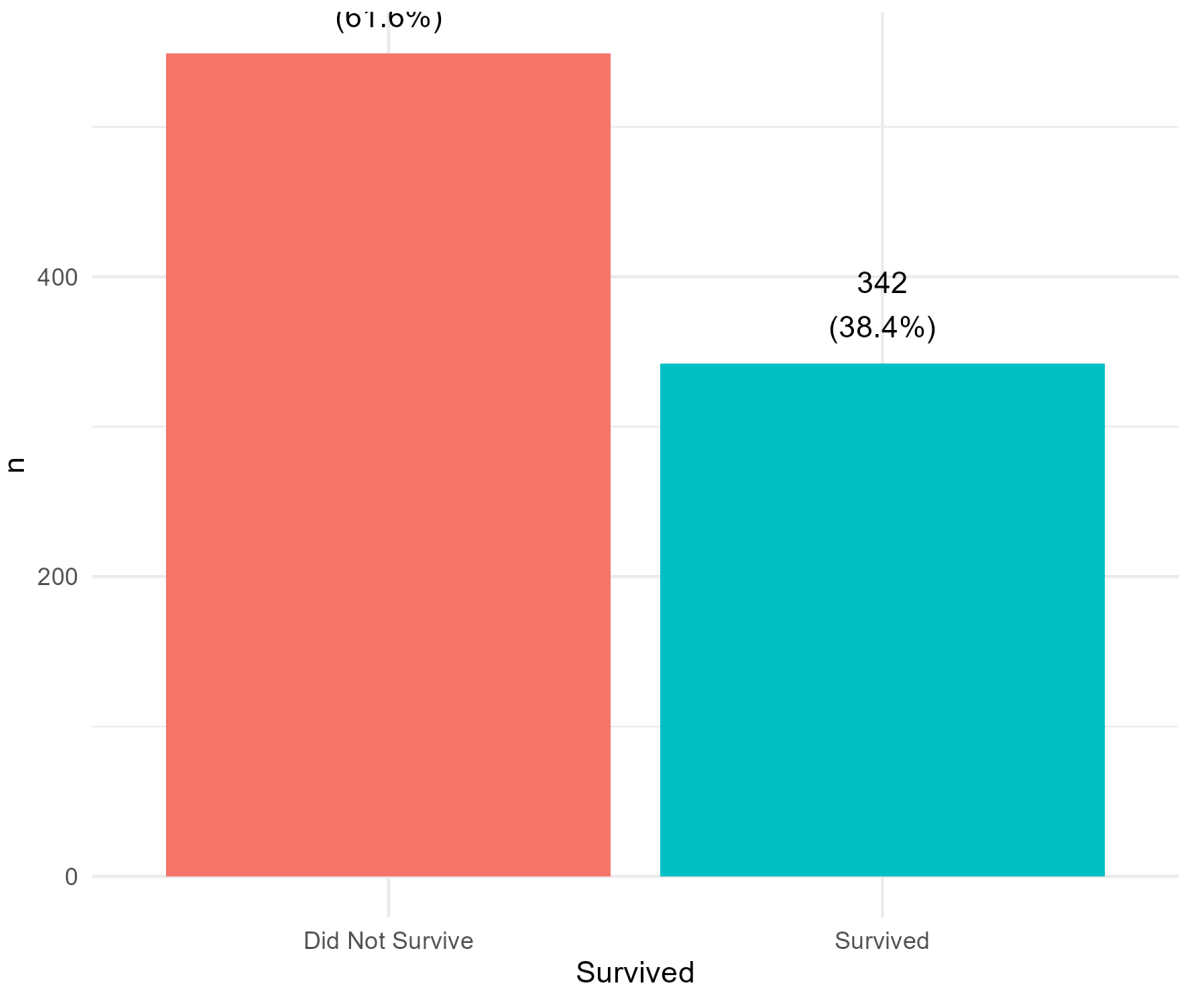
Part 1 — Titanic Dataset



Plot 01 — Missing Values Bar Chart

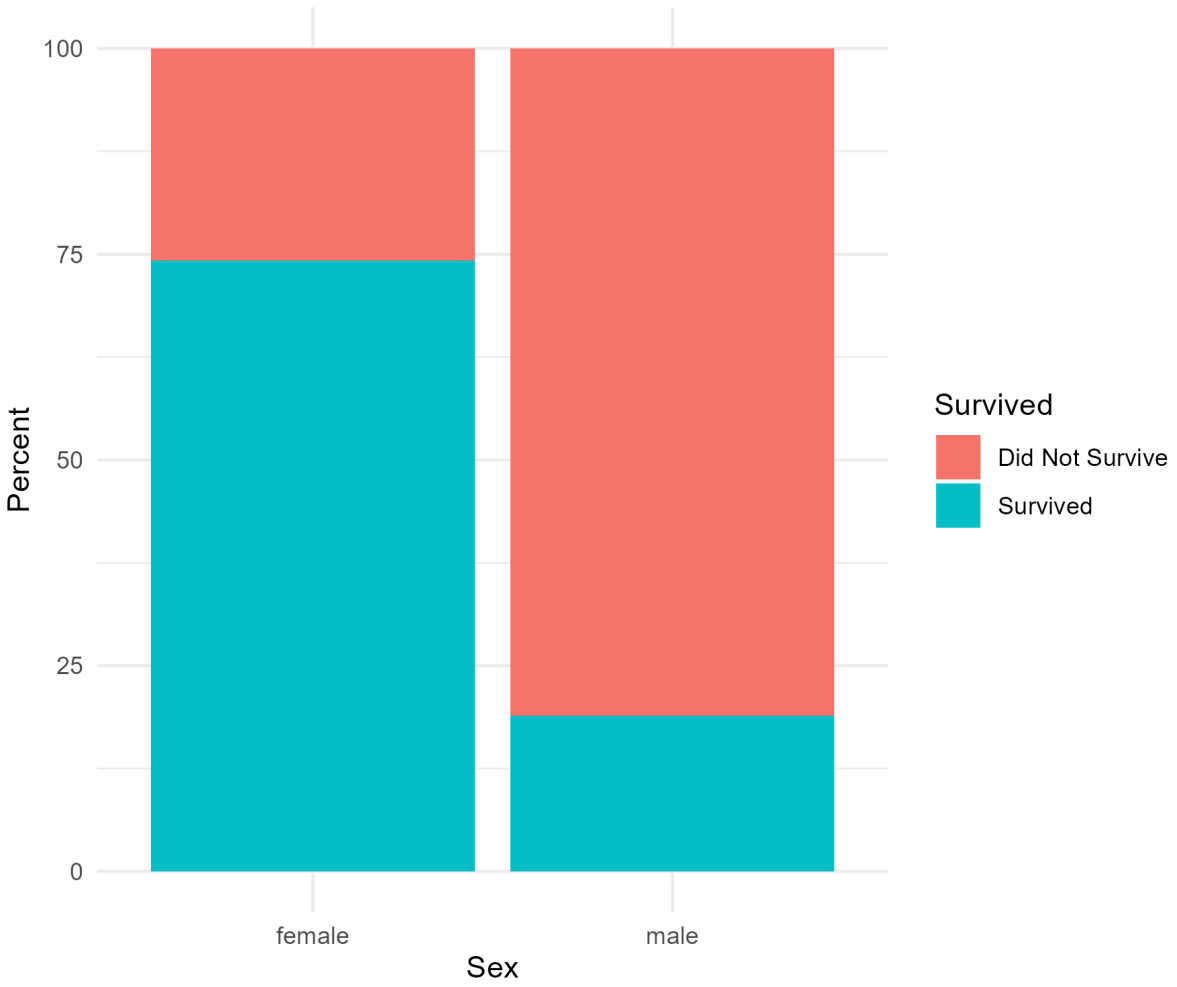
- Cabin had the worst data quality with **687 missing values (77.1%)** — more than three-quarters of the column was empty, making it nearly unusable for analysis
- Age had **177 missing values (19.9%)** — significant enough to skew results if not handled, so median imputation was the right choice
- Embarked had only **2 missing values (0.2%)**

Plot 02 — Overall Survival Bar Chart



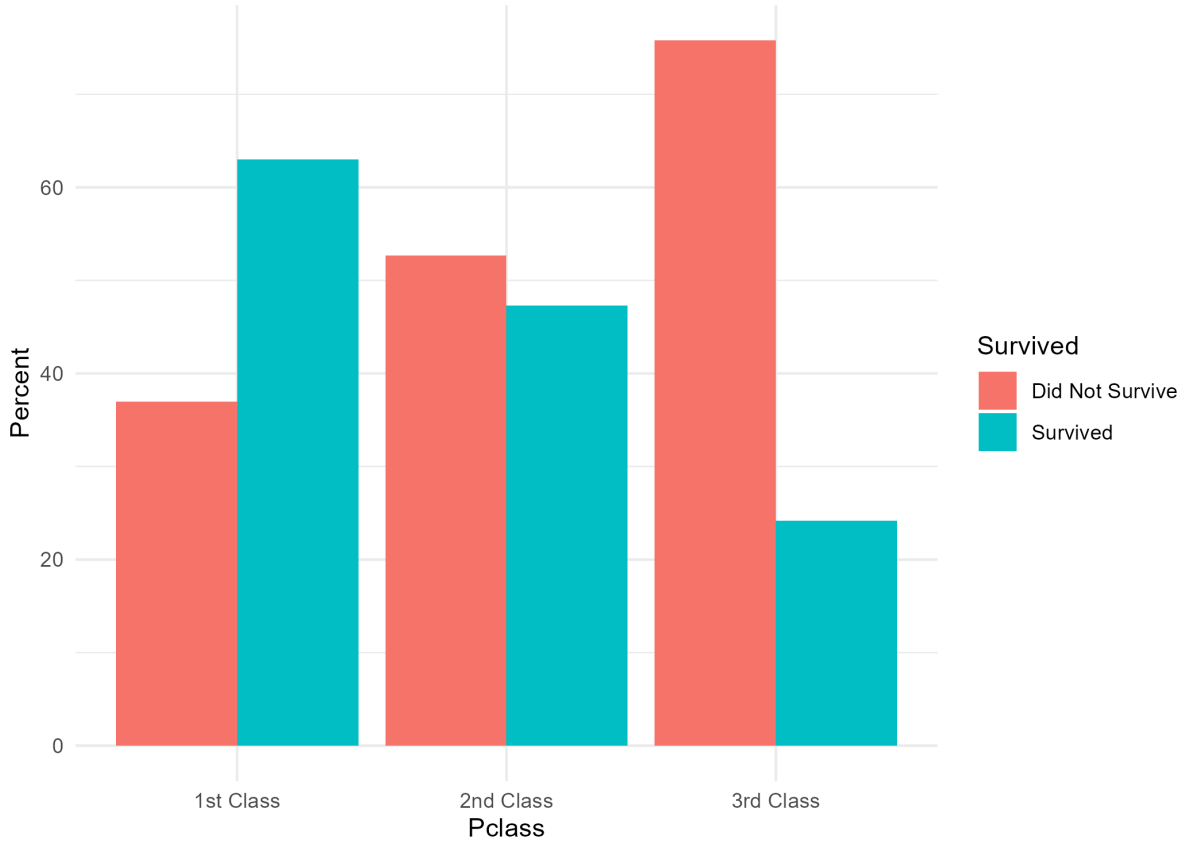
- **549 passengers (61.6%) did not survive** vs only 342 (38.4%) who did
- The chart clearly shows death outnumbered survival by nearly **2:1**

Plot 03 — Survival by Sex Stacked Bar



- **Females: ~74% survived** (large teal section at the bottom)
- **Males: only ~19% survived** (small teal section)

Plot 04 — Survival by Class Bar Chart



- Still shows the clearly: 1st class teal bar towers over 3rd class teal bar

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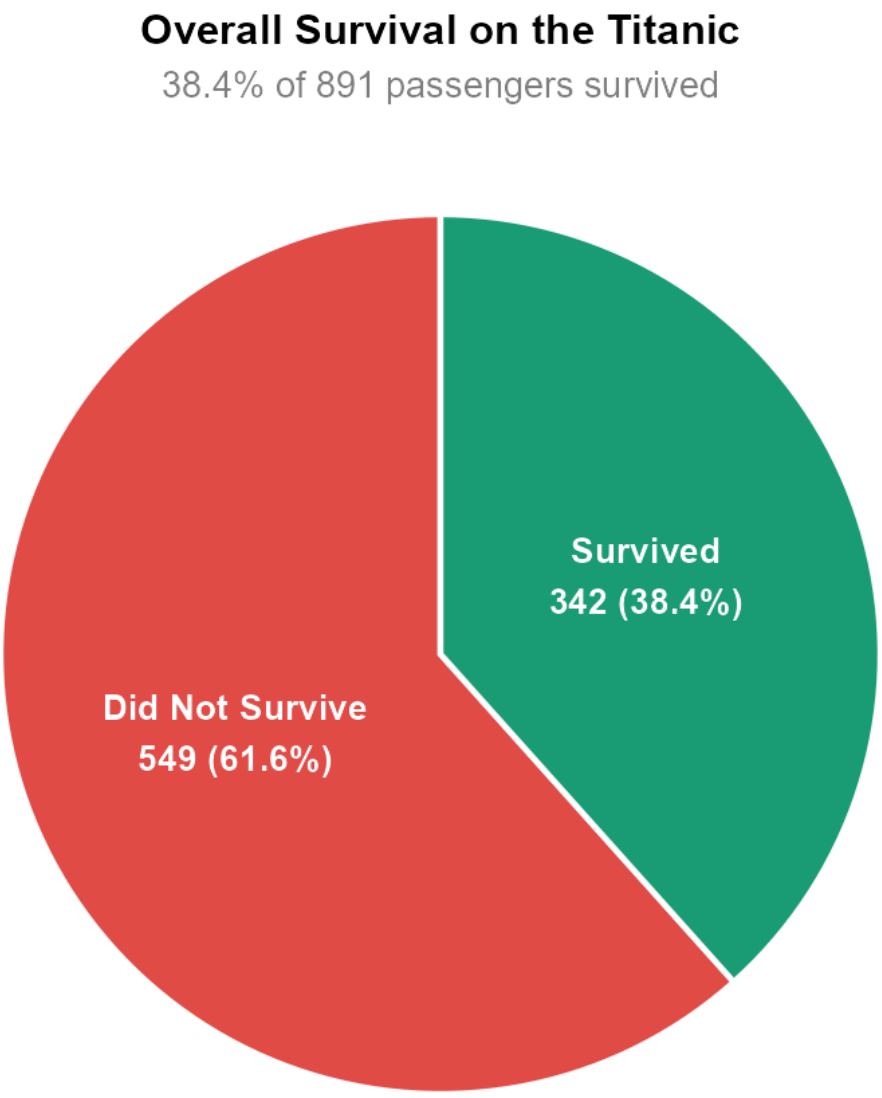
T1 — Pie Chart



- **Did Not Survive 549 (61.6%)** in red, **Survived 342 (38.4%)** in green

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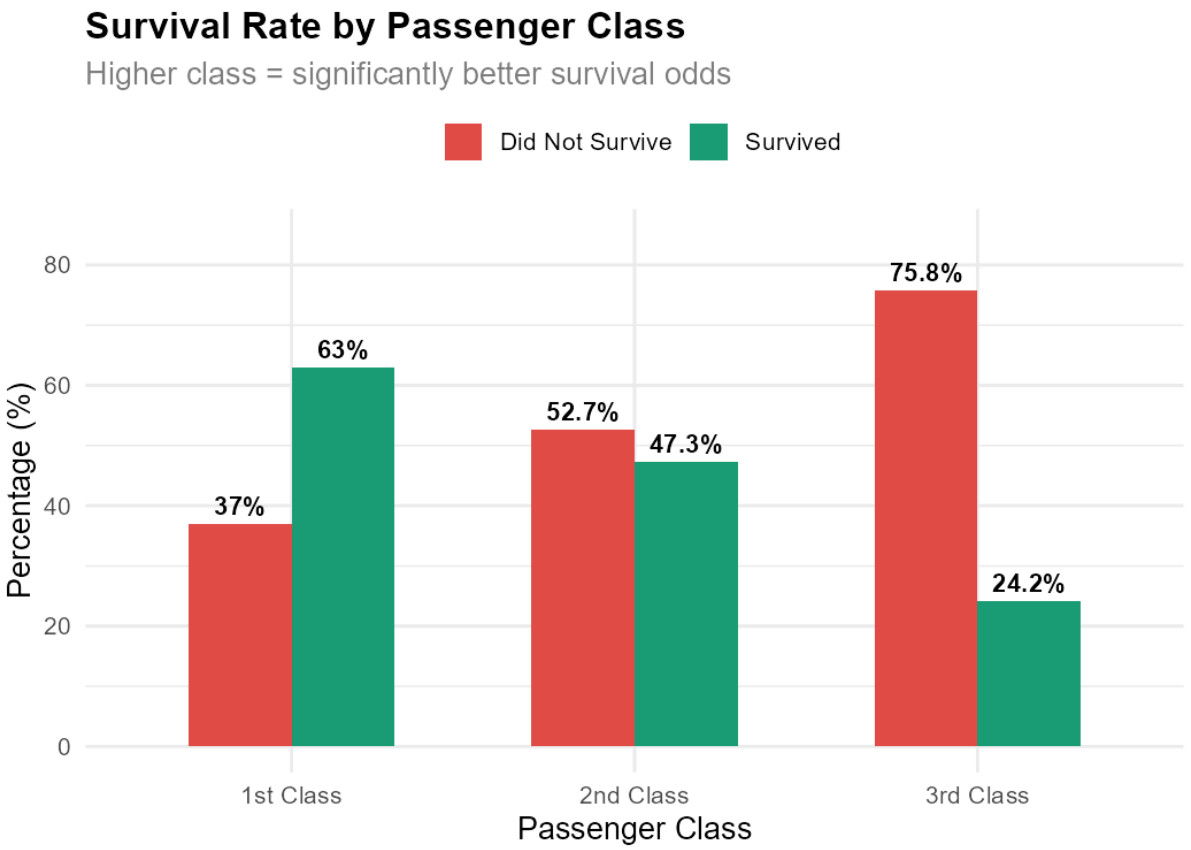
T2 — Bar Chart (Survival by Class, improved)



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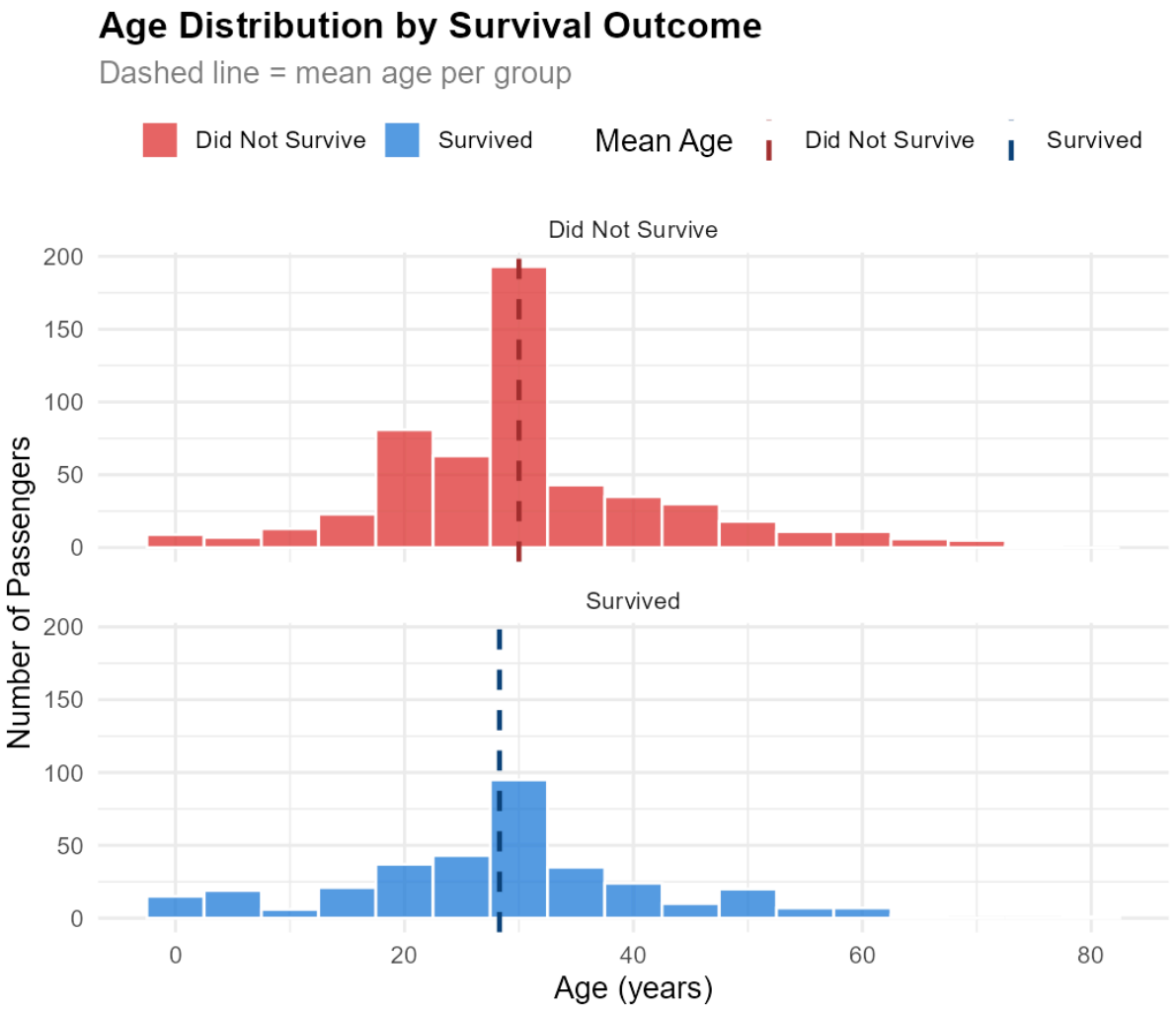
- **1st Class**: 63% survived vs 37% did not — the only class where survivors outnumber deaths
- **2nd Class**: almost even split — 52.7% died vs 47.3% survived — a coin flip
- **3rd Class**: 75.8% did not survive vs only 24.2% survived — the most tragic outcome
- Conclusion: **wealth/class was a strong predictor of survival**

T3 — Histogram (Age Distribution by Survival)



- Both groups show a large peak at the **25–30 age bin** — this is partly artificial because missing ages were filled with the median (28), creating that spike
- The dashed lines show mean age of **30 for non-survivors** vs **28.3 for survivors**
- Both distributions look similar overall, meaning **age alone was not a strong predictor**
- Small bars at age 0–5 are visible in both panels, confirming some young children were on board

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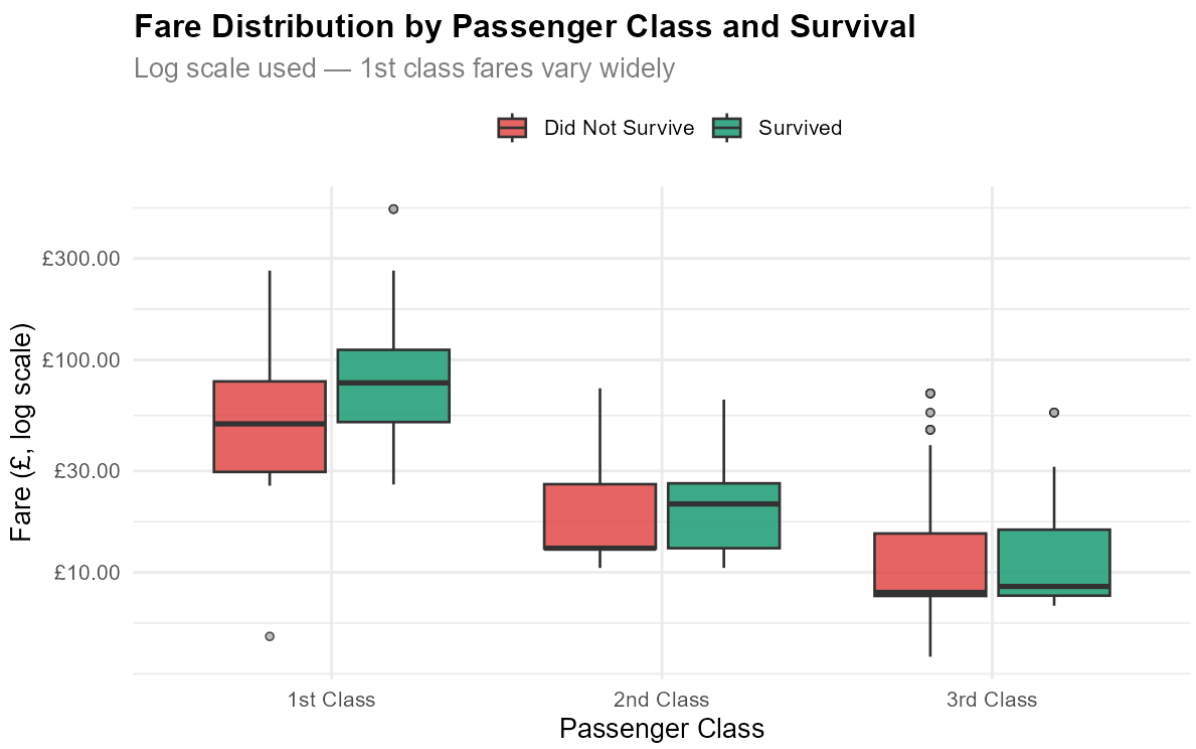
T4 — Box Plot (Fare by Class and Survival)



- **1st Class**: survivors (green box) have a noticeably higher median fare than non-survivors (red box) — paying more within 1st class still gave better odds
- **2nd Class**: the two boxes are very close — fare within 2nd class did not strongly separate survivors from non-survivors
- **3rd Class**: both boxes are tightly clustered near the bottom (£8–£15), and the difference between survivors and non-survivors is minimal — in 3rd class, fare did not matter much; their outcome was largely determined by class itself

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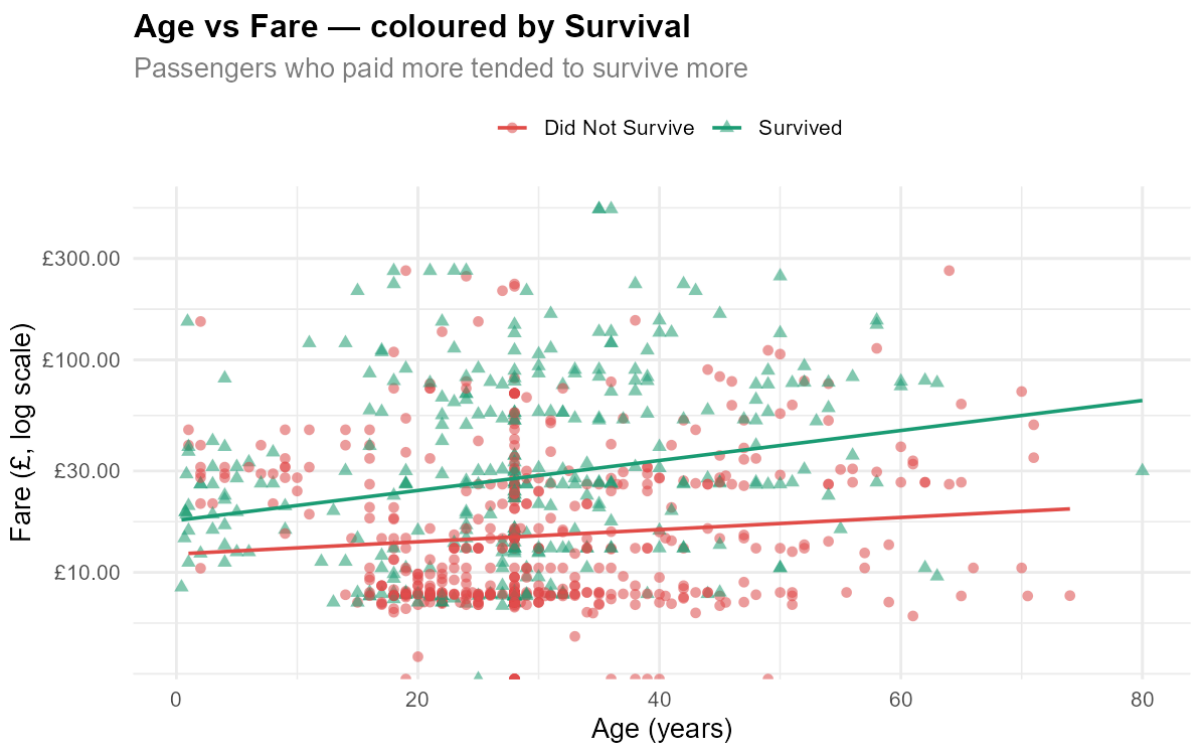
T5 — Scatter Plot (Age vs Fare coloured by Survival)



- The **green triangles (Survived) cluster noticeably higher** on the fare axis compared to red circles (Did Not Survive)
- The green trend line sits consistently above the red trend line across all ages — confirming that **higher fare = better survival odds** regardless of age
- Both trend lines slope upward slightly with age — older passengers tended to pay more, likely because they were wealthier
- A dense cluster of red dots at the bottom left (young passengers, low fare) represents the bulk of 3rd class male non-survivors

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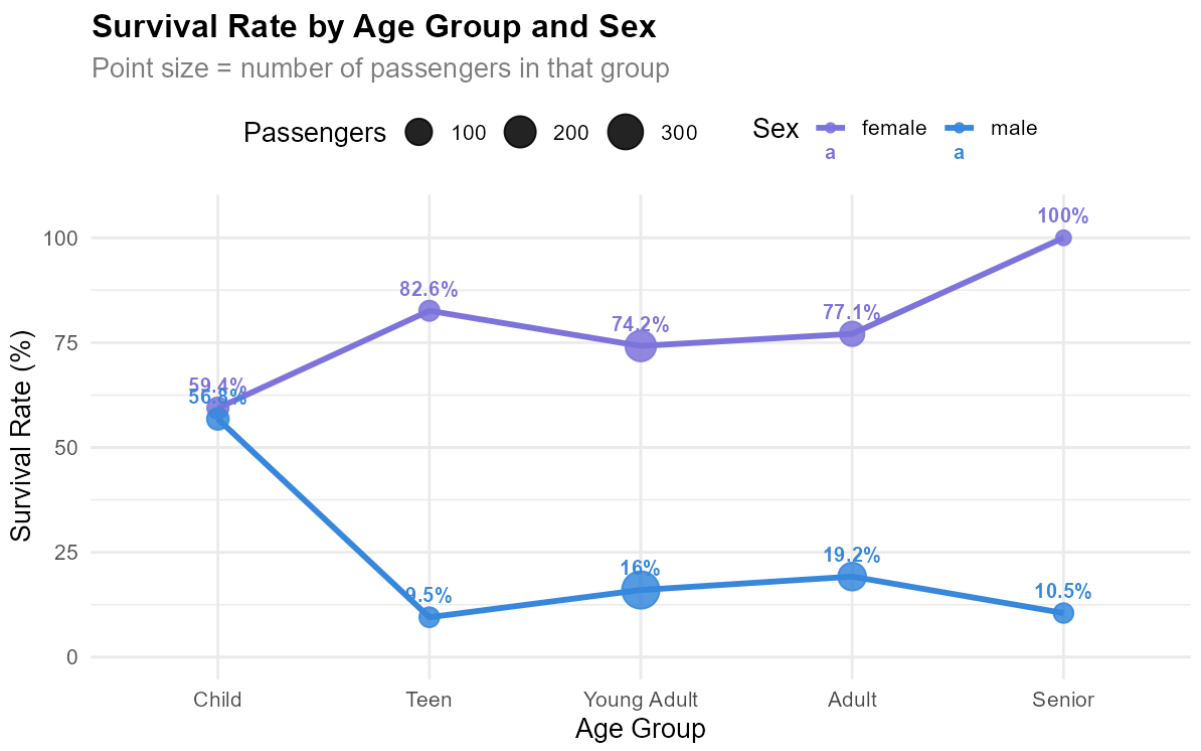
T6 — Line Graph (Survival Rate by Age Group and Sex)



- **Female line (purple) stays consistently high**: Child 59.4% → Teen 82.6% → Young Adult 74.2% → Adult 77.1% → Senior 100%
- **Male line (blue) drops sharply after childhood**: Child 56.8% → Teen 9.5% → Young Adult 16% → Adult 19.2% → Senior 10.5%
- At the **Child group**, male and female survival rates are almost equal (56.8% vs 59.4%) — this confirms young boys were also prioritised during evacuation
- The gap between male and female survival **widens dramatically from Teen onwards** — once past childhood, being male was a severe disadvantage
- Point sizes reflect group sizes — the largest points are Young Adult males (375 passengers) and Young Adult females (186) — these were the dominant age groups on the ship
- The Senior female 100% result (3 passengers) should be interpreted cautiously due to the very small sample size

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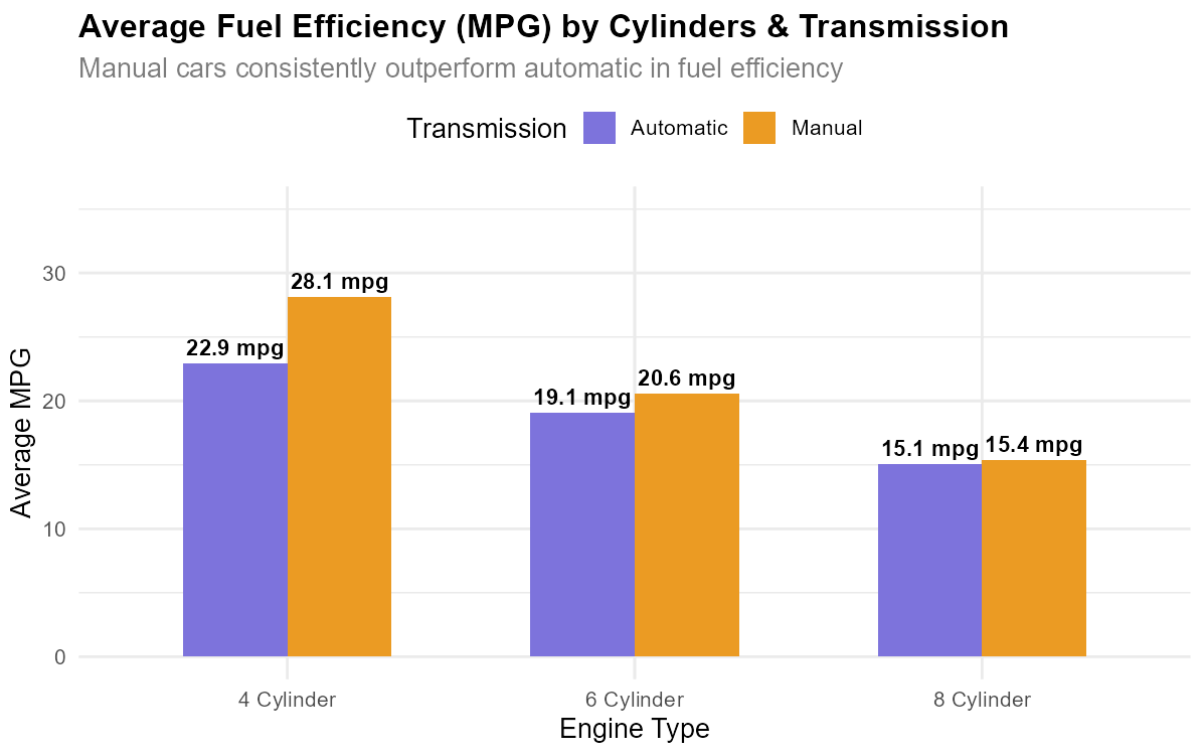
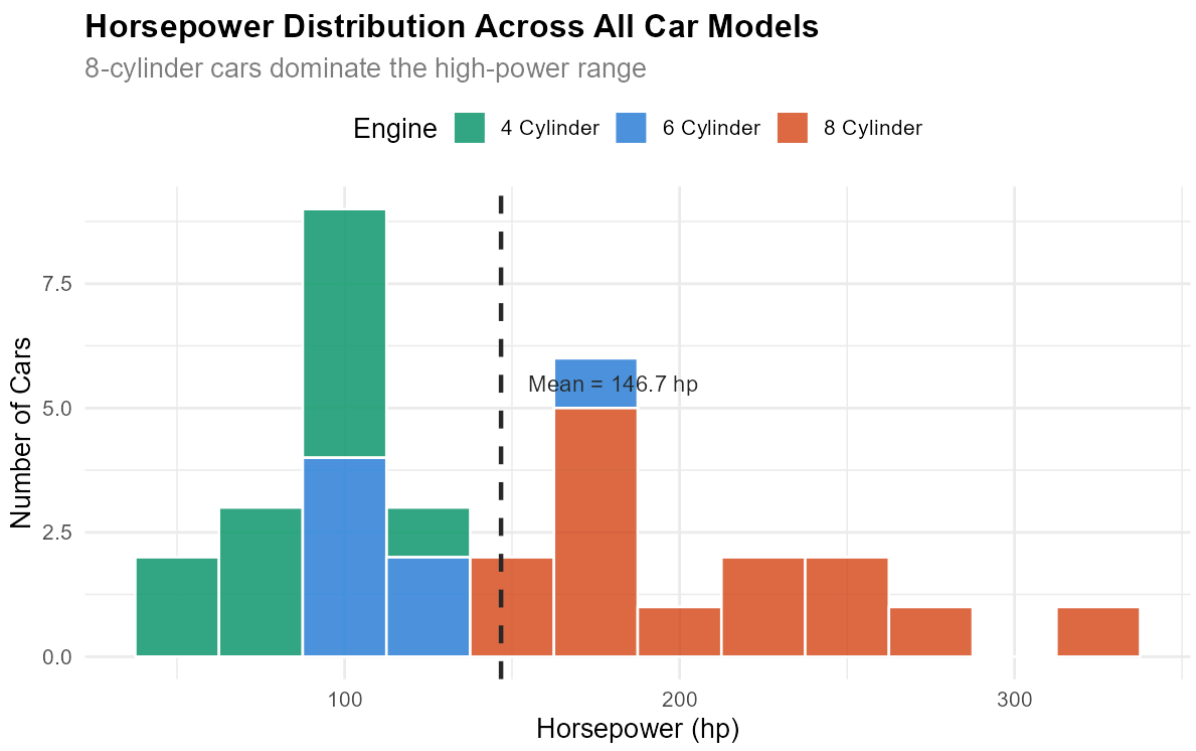
Part 2 — mtcars Dataset



M2 — Bar Chart (MPG by Cylinder and Transmission)

- **4-cylinder manual: 28.1 mpg** — best fuel efficiency in the entire dataset
- **8-cylinder automatic: 15.1 mpg** — worst, nearly half the efficiency of a 4-cyl manual
- Manual beats automatic in **all three cylinder categories**, confirming transmission type matters
- The gap between manual and automatic is largest for 4-cylinder cars (28.1 vs 22.9 = **5.2 mpg difference**) but almost disappears for 8-cylinder (15.4 vs 15.1 = only **0.3 mpg difference**)

M3 — Histogram (Horsepower Distribution)



- Right-skewed distribution — most cars cluster in the **75–150 hp range**, with a long tail extending to 335 hp
- The mean line at **146.7 hp** sits to the right of the tallest bar, confirming the right skew
- **4-cylinder (green)** cars dominate the 75–125 hp range
- **6-cylinder (blue)** cars fill the 100–150 hp overlap zone
- **8-cylinder (orange)** cars take over from 150 hp onwards exclusively — no 4 or 6 cylinder car exceeds 150 hp in this dataset

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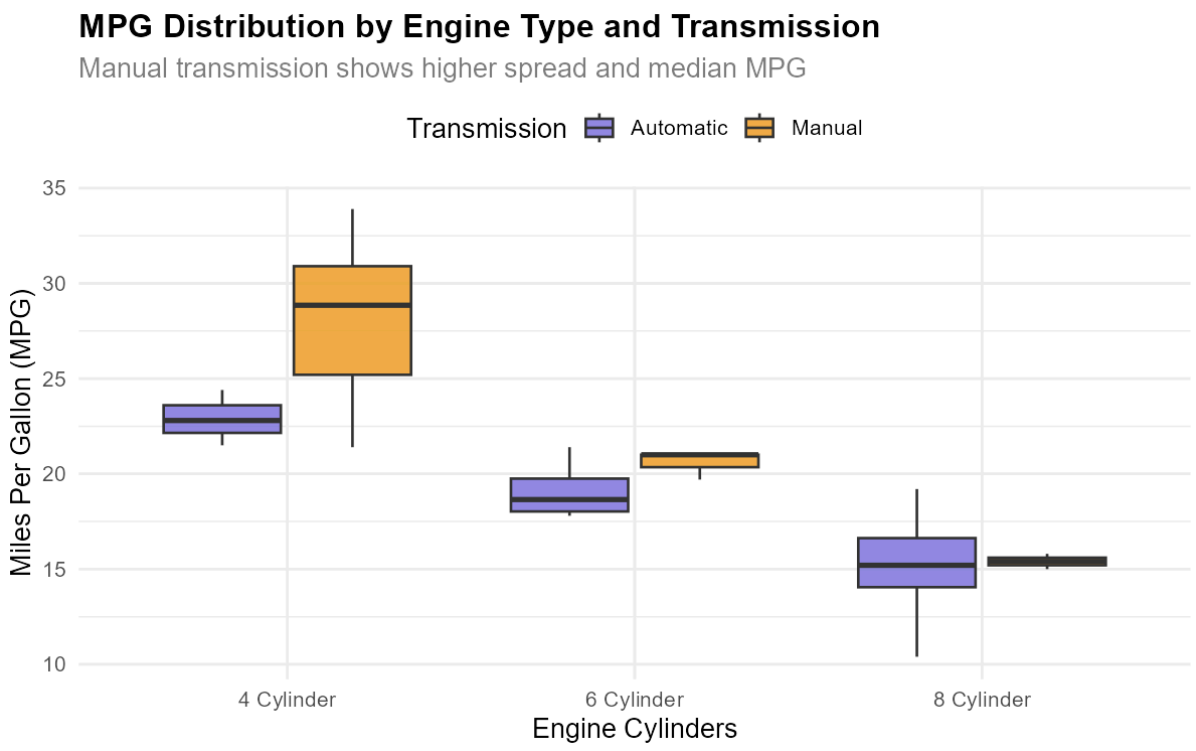
M4 — Box Plot (MPG by Cylinder and Transmission)



- **4-cylinder manual (yellow/orange box)** has the widest spread — ranging from about 21 to 34 mpg, with median around 29 mpg — the most variable and best performing group
- **4-cylinder automatic (purple box)** is tightly clustered around 22–24 mpg — consistent but lower than manual
- **6-cylinder manual** has only a flat line (very thin box) — this is because there are only **3 manual 6-cylinder cars** in the dataset, so there is almost no spread
- **8-cylinder manual** also appears as a very flat line for the same reason — only **2 cars**
- **8-cylinder automatic** shows the most outlier-free, compact distribution — all sitting between 10–19 mpg, confirming these are uniformly poor on fuel economy

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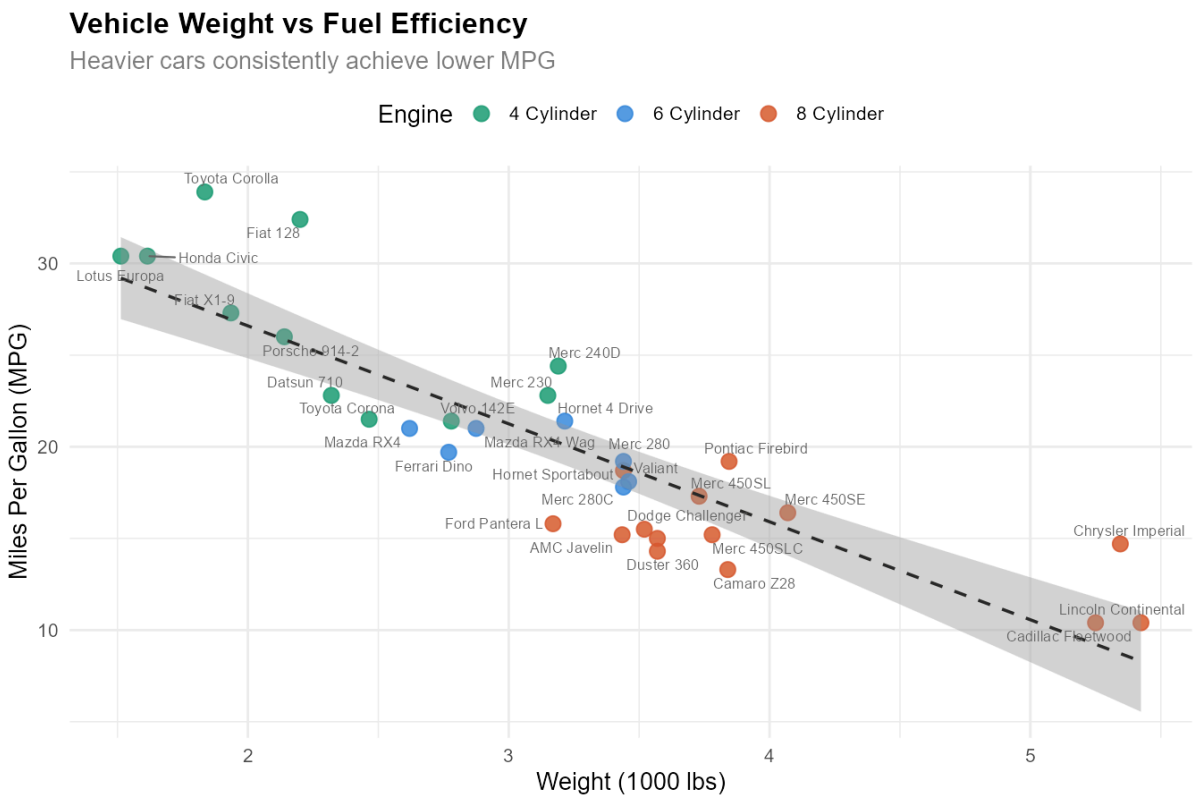
M5 — Scatter Plot (Weight vs MPG with car labels)



- **Strong negative correlation** — the dashed regression line slopes clearly downward from top-left to bottom-right
- **Top-left cluster (light, high MPG)**: Toyota Corolla (33.9 mpg, 1.835k lbs), Fiat 128, Honda Civic, Lotus Europa — all 4-cylinder green dots
- **Bottom-right cluster (heavy, low MPG)**: Cadillac Fleetwood, Lincoln Continental, Chrysler Imperial — all 8-cylinder orange dots weighing over 5,000 lbs
- All **6-cylinder (blue)** dots sit in the middle band — moderate weight, moderate MPG